📒 General Ledger (GL) Transaction Engine — Deep HLD Design

> ***Document Focus:*** *GL Transaction Engine Architecture · Auto-Posting Rules · Dynamic Module Integration*
> ***Project:*** *GTR Service ERP — Multi-Tenant SaaS Platform*
> ***Reference:*** *Accounting Module HLD · Accounting Module (3).docx*
> ***Style:*** *HLD Design Only — No Code*

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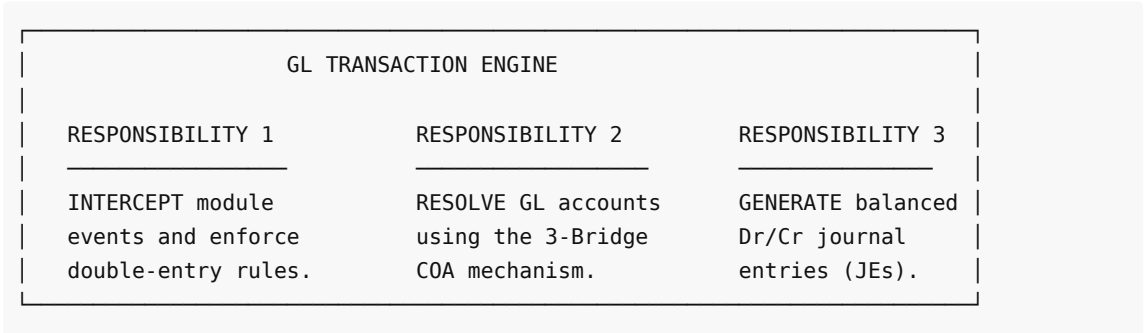
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1. What is the GL Transaction Engine?

The **GL Transaction Engine** is the core operational heart of the Accounting Module. While the Chart of Accounts (COA) Engine defines *where* money can go, the GL Transaction Engine handles *how* and *when* money moves.

Every single business event across the ERP—whether it's a warehouse transfer, a sales invoice, a payroll run, or a depreciation calculation—must pass through the GL Transaction Engine to be recorded in the system via **Double-Entry Bookkeeping**.

Core Responsibilities

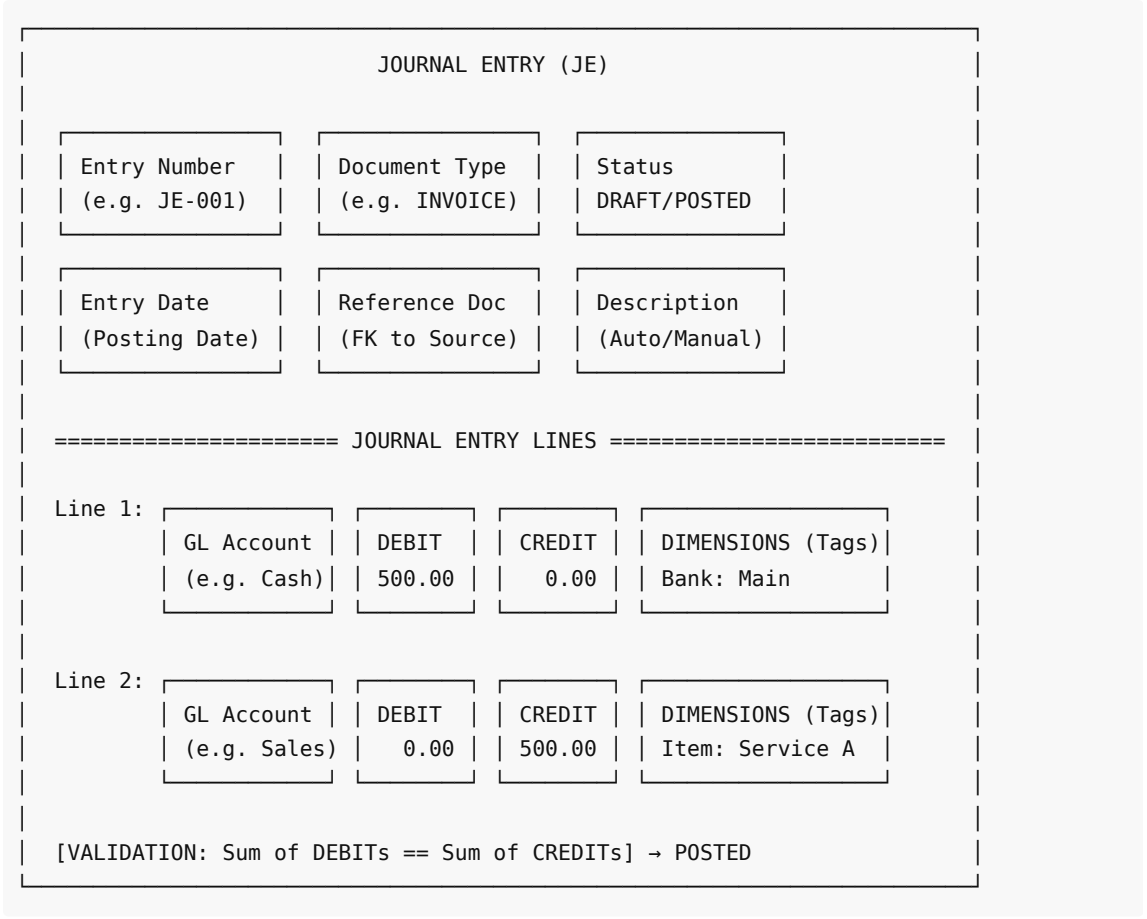


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2. GL Transaction Anatomy

A transaction in the GL Engine is represented by a **Journal Entry**. The engine enforces the universal accounting rule: **Total Debits must equal Total Credits**.

2.1 The Journal Entry Structure



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3. Auto-Posting Rules Engine

The true power of the GL Transaction Engine lies in its **Auto-Posting Rules**. Users in operational modules (like Sales or Warehouse) do not need to know debits and credits. They just click "Post" on their document, and the GL Engine handles the rest.

3.1 Document Parsing Flow

When the `accounts_service` receives a posting request from any module, it runs through the Auto-Posting Rules Engine:

1. **Document Identification**: Is this a `PURCHASE_INVOICE`, `SALES_RECEIPT`, `INVENTORY_TRANSFER`, or `DEPRECIATION_RUN`?
2. **Context Extraction**: Who is the partner (Company)? What are the items (Products/Services)? Which warehouse? What is the VAT rate?
3. **Bridge Lookups**:
   - Query `ItemGLAccount` (Bridge 1) to find Item-specific accounts (e.g., Goods, COGS, Sales Revenue).
   - Query `CompanyGLAccount` (Bridge 2) to find Partner-specific accounts (e.g., Trade Receivables, Payables, Bank).
   - Query `VAT Rates` to find Tax accounts (e.g., Input/Output VAT).
4. **Drafting the JE**: Assemble the Journal Entry lines based on the predefined accounting rules for that document type.

5. **Balancing & Posting**: Check `Dr == Cr`. If balanced, change status to `POSTED` and lock the transaction.

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4. Dynamic Integration with Existing Modules

Here is a deep dive into the specific Auto-Posting rules for each major module currently in the system.

4.1 Purchases Module Auto-Posting

When an **Invoice Received** is posted by the procurement team:

```
BUSINESS EVENT: Purchase Invoice #INV-1001 for 10 Laptops + 5% VAT
_____________________________________________________________________

Rule: Increase Inventory (Asset), Record Input VAT (Asset), Increase Liability
(Payables).

Generated JE:
    Dr: Goods (BS Inventories)                [Derived from ItemGLAccount]
    Dr: Current VAT Receivable (BS Input VAT) [Derived from VAT Type]
    Cr: Current Trade Payables (BS Liability) [Derived from CompanyGLAccount]
```

When a **Purchase Payment** is made via Bank:

```
BUSINESS EVENT: Payment of AED 5,250 via Bank Transfer
_____________________________________________________________________

Rule: Decrease Liability (Payables), Decrease Asset (Bank).

Generated JE:
    Dr: Current Trade Payables
    Cr: Bank Account
```

4.2 Sales Module Auto-Posting

When a **Customer Invoice** is posted:

```
BUSINESS EVENT: Sales Invoice #SAL-2001 for Services + 5% VAT
_____________________________________________________________________

Rule: Increase Asset (Receivables), Recognize Income, Record Output VAT (Liability).

Generated JE:
    Dr: Trade Receivables (BS Asset)
    Cr: Sales of Services (P&L Income)
    Cr: Current VAT Payable (BS Liability)
```

When a **Deferred Revenue** invoice is posted (e.g., 1-year software subscription):

```
BUSINESS EVENT: Invoice for 1-Year Subscription
_____________________________________________________________________

Rule: Record Receivables, Defer Revenue to Liability.

Generated JE (On Invoice):
    Dr: Trade Receivables
```

```
    Cr: Deferred Revenue (BS Liability)

Generated JE (On Monthly Revenue Recognition via Engine):
    Dr: Deferred Revenue
    Cr: Sales of Services (P&L Income)
```

4.3 Warehouse Module Auto-Posting

When an **Inventory Transfer** is posted:

```
BUSINESS EVENT: Transfer 50 Units from Main Warehouse to Branch
──────────────────────────────────────────────────────────────────

Rule: Move value between dimensions. Net financial impact is zero, but Trial Balance
updates per warehouse.

Generated JE:
    Dr: Goods [Dimension: Branch Warehouse]
    Cr: Goods [Dimension: Main Warehouse]
```

When an **Inventory Write-Off** is posted (e.g., damaged goods):

```
BUSINESS EVENT: Write-off 5 Damaged Units
──────────────────────────────────────────────────────────────

Rule: Recognize Expense, Reduce Asset.

Generated JE:
    Dr: Other Operating Expenses (P&L)
    Cr: Goods (BS Inventories)
```

4.4 Tax / VAT Engine

```
BUSINESS EVENT: VAT Settlement Payment to FTA
──────────────────────────────────────────────────────────────

Rule: Clear Payable Liability, Reduce Cash.

Generated JE:
    Dr: Current VAT Payable
    Cr: Bank Account
```

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5. Upcoming Module Integration Design

The dynamic design of the GL Engine means that **new modules do not require structural changes to the accounting core**. They simply map to existing Document Types or create new ones, utilizing the existing Three-Bridge COA resolution.

5.1 Applicant Tracking System (ATS) Integration

Scenario: Charging clients for successful candidate placements.

```
MODULE EVENT: ATS Candidate Placed (B2B Recruitment Service)
──────────────────────────────────────────────────────────────
```

```
Integration Flow:
1. ATS triggers "Generate Invoice" via accounts_service.
2. Item = "Recruitment Fee" (Service).
3. GL Resolution:
   - Dr: Trade Receivables (CompanyGLAccount)
   - Cr: Sales of Services (ItemGLAccount for "Recruitment Fee")
   - Cr: Current VAT Payable

Dimension Tracking:
- Contract ID / Job Order ID can be passed as a dimension to track profitability per
recruitment campaign.
```

5.2 Supply & Tendering Module

Scenario: Supplier deposits / Tender Bonds.

```
MODULE EVENT: Tender Bond Received from Sub-Contractor
──────────────────────────────────────────────────────────────────────

Integration Flow:
1. Tendering module posts a Receipt.
2. GL Resolution:
   - Dr: Bank Account / Cash
   - Cr: Prepayments from Suppliers (or dedicated "Tender Deposits Payable" liability
GL)
```

5.3 Trip Management / Fleet Module

Scenario: Driver logs fuel expenses and trip revenue.

```
MODULE EVENT A: Fuel Expense Logged
──────────────────────────────────────────────────────────────────────

- Dr: Fuel Expense (P&L) [Dimension: Vehicle ID]
- Cr: Cash on Hand / Driver Advance (BS)

MODULE EVENT B: Trip Completed & Billed
──────────────────────────────────────────────────────────────────────

- Dr: Trade Receivables
- Cr: Sales of Services (Transport Income)
```

5.4 POS (Point of Sale) Integration

Scenario: Daily retail store sales.

```
MODULE EVENT: POS Daily Shift Close
──────────────────────────────────────────────────────────────────────

Integration Flow:
Instead of posting individual receipts, POS sends a consolidated daily shift report.
- Dr: Cash on Hand (for cash sales)
- Dr: Bank Account (for credit card machine settlements)
- Cr: Sales of Goods (P&L)
- Cr: Current VAT Payable
```

```
- Dr: COS (Cost of Sales)
- Cr: Goods (Inventory Reduction)
```

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6. GL to Financial Statements Flow

Once Journal Entries are safely inside the `journal_entries` and `journal_entry_lines` tables, the Financial Reporting Engine takes over to generate statements in real-time.

6.1 Trial Balance Calculation

The Trial Balance is the foundation of all reports. It calculates balances dynamically:

1. **Opening Balance:** Sum of all `Dr - Cr` lines for the GL account *prior* to the selected date range.
2. **Turnovers:**
   - Debit Turnover: Sum of `Dr` lines *during* the date range.
   - Credit Turnover: Sum of `Cr` lines *during* the date range.
3. **Closing Balance:** `Opening Balance + Debit Turnover - Credit Turnover`.

6.2 Posting Reports (Document-Level Audit)

Every document (Invoice, Payment, Transfer) has a `Posting Report` button. This queries the `journal_entries` table filtering by `referenceDocument = <doc_id>` to show exactly what debits and credits were automatically generated, ensuring full auditability.

6.3 Final Generation

```
GL TRANSACTIONS
      │
      ▼
TRIAL BALANCE (Groups balances by GL Account & Dimensions)
      │
      ├─► BALANCE SHEET (Filters by Section = 'BS *')
      │
      ├─► INCOME STATEMENT (Filters by Section = 'P&L *')
      │
      └─► STATEMENT OF ACCOUNT (Filters by GL + Company Dimension)
```